

$$\text{Self Attention } (Q, K, V) = \frac{\text{Softmax}(QK^T)}{\sqrt{d_k}} V$$

$$Q \in \mathbb{R}^{N \times d}$$

N - seq. length

$$K \in \mathbb{R}^{N \times d}$$

d - Head dimension

$$V \in \mathbb{R}^{N \times d}$$

~~Input vector size~~

d_{model} - size of embedding of each word.

Since, multi-head attention process "slices" of the embedding vector, $d = \frac{d_{\text{model}}}{h}$ ($h = \#$ of Attn heads)

If O is the o/p, then $O \in \mathbb{R}^{N \times d}$

$$S = QK^T$$

$$P = \text{Softmax}(S)$$

$$O = P \cdot V$$

$$\boxed{\text{Softmax}(x_i) = \frac{e^{x_i - m(x)}}{\sum_j e^{x_j - m(x)}}$$

$$m(x) = \max(x_1, \dots, x_n)$$